

DBMS COURSE PROJECT

Cinema Screening and Ticket Booking System

CinemaDB | MySQL 8.0

Team Members and Roll Numbers

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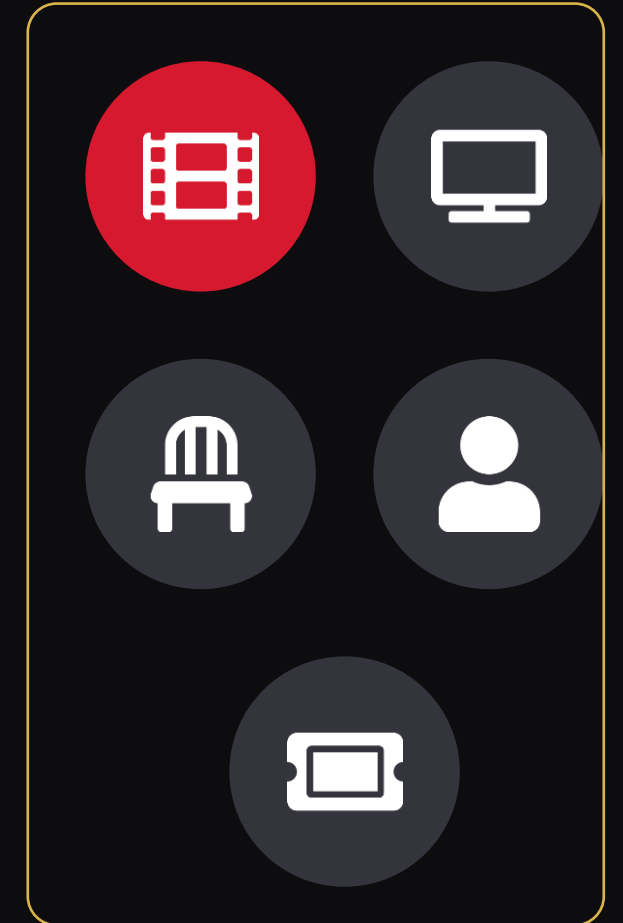
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Problem Description

The Problem

- Manual handling of movies, screens, seats, customers and bookings causes redundancy and inconsistency
- A centralized relational database stores and retrieves data efficiently
- Bookings link customers, screenings and seats

Intended Users

-  Cinema administrators
-  Booking staff
-  Customers

Objectives

- 1 Manage movie and screening information
- 2 Manage customers and seats
- 3 Store ticket bookings
- 4 Prevent duplicate seat booking for the same screening
- 5 Perform SQL queries and generate useful information
- 6 Provide a user interface for database operations

System Scope and Features



Movie Management

Store title, genre, language, duration, rating



Screen Management

Maintain screens and their capacity



Seat Management

Seats for each screen with seat type



Customer Management

Customer name, phone and email



Screening Management

Schedule movies on screens by date and time



Booking Management

Record tickets, amount and status



Reports / Queries

JOIN and aggregate SQL results



Dashboard / UI

Insert, delete and view records

ER Diagram

PK Primary Key **FK** Foreign Key **1 : M** One-to-Many



Relational Schema

PK Primary Key **FK** Foreign Key



Movie (movie_id PK, title, genre, language, duration, rating)



Screen (screen_id PK, screen_name, capacity)



Customer (customer_id PK, name, phone, email)



Seat (seat_id PK, screen_id FK, seat_number, seat_type)



Screening (screening_id PK, movie_id FK, screen_id FK, show_date, show_time)



Booking (booking_id PK, customer_id FK, screening_id FK, seat_id FK, booking_date, amount, status)

Normalization up to 3NF

1NF

First Normal Form

- All attributes hold atomic values
- No repeating groups



2NF

Second Normal Form

- Satisfies 1NF
- Non-key attributes depend on the complete primary key



3NF

Third Normal Form

- Satisfies 2NF
- No transitive dependency
- Non-key attributes depend only on the primary key

CinemaDB is designed up to 3NF to reduce redundancy and update anomalies.

Database Implementation and SQL

```
CREATE DATABASE CinemaDB;
USE CinemaDB;

CREATE TABLE Booking (
    booking_id    INT PRIMARY KEY AUTO_INCREMENT,
    customer_id   INT NOT NULL,
    screening_id  INT NOT NULL,
    seat_id       INT NOT NULL,
    booking_date  DATE NOT NULL,
    amount        DECIMAL(8,2) NOT NULL,
    status        VARCHAR(20) NOT NULL,
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id),
    FOREIGN KEY (screening_id) REFERENCES Screening(screening_id),
    FOREIGN KEY (seat_id) REFERENCES Seat(seat_id),
    UNIQUE (screening_id, seat_id)
);
```

6

tables created

5+

records in each table

FK

referential integrity
maintained

PRIMARY KEY | FOREIGN KEY | NOT NULL | UNIQUE

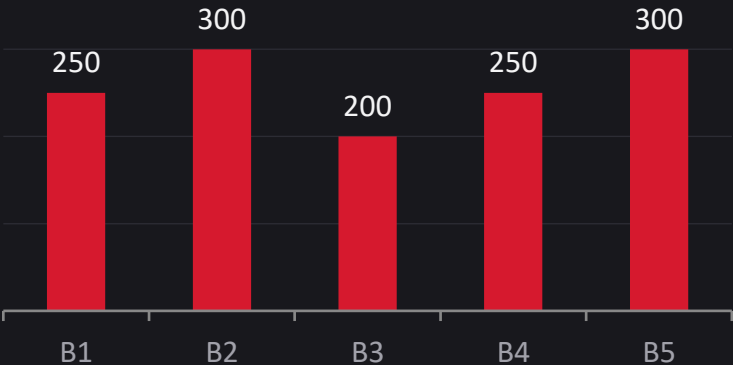
SQL Queries and Results

#	Query	Technique
1	Movie and screening details	JOIN
2	Complete booking details	Multiple JOINS
3	Bookings for each movie	COUNT + GROUP BY
4	Confirmed booking amount	SUM
5	Average ticket price	AVG
6	Highest ticket price	MAX
7	Customers who booked tickets	JOIN
8	Movies with rating above 8	WHERE
9	Premium seats	WHERE
10	Confirmed bookings	WHERE

TOTAL CONFIRMED REVENUE

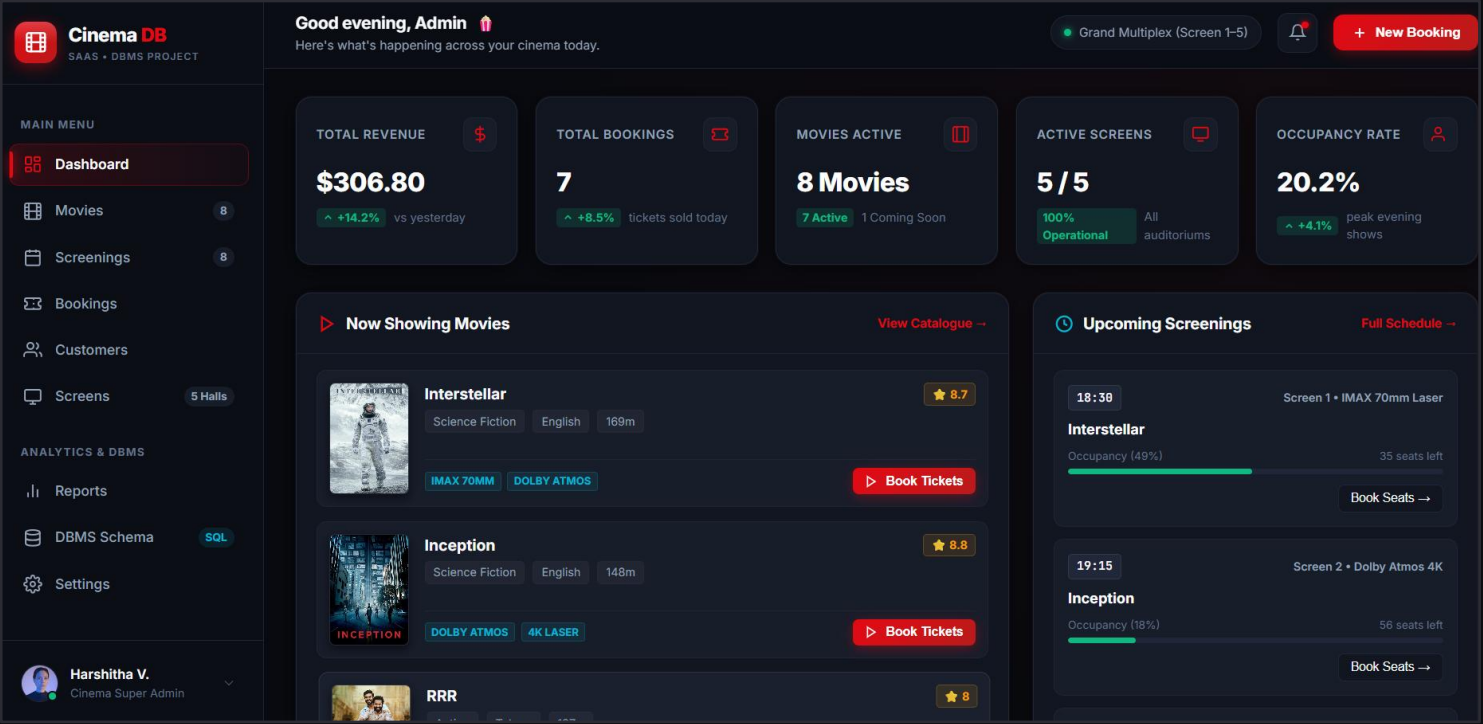
₹1,300

Ticket amounts (₹)

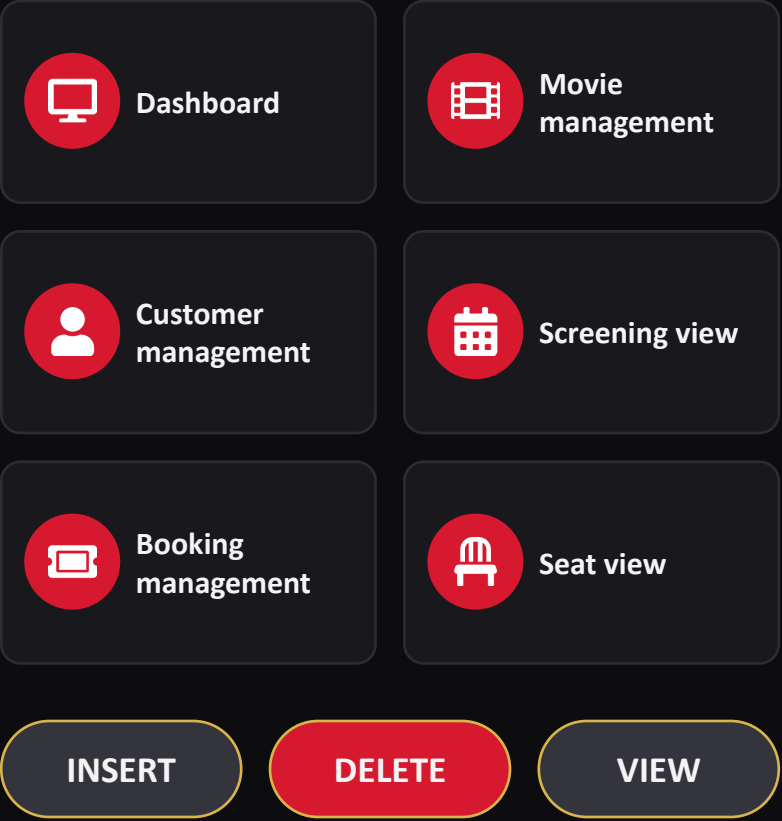


AVG ₹260 MAX ₹300

User Interface / Dashboard



UI design prototype (dashboard mockup, sample data)



Changes made in the UI are reflected in the MySQL CinemaDB.

Testing, Conclusion and Future Scope

Testing

- ✓ Database connection
- ✓ Tables created
- ✓ Records inserted
- ✓ SELECT queries verified
- ✓ JOIN queries verified
- ✓ INSERT operation
- ✓ DELETE operation
- ✓ Foreign-key constraints
- ✓ Duplicate seat booking constraint

Conclusion

The system provides a structured relational database for managing movies, screens, seats, customers, screenings and bookings.

Future Enhancements

Online payment integration

Customer login

Real-time seat selection

Email/SMS confirmation

Advanced analytics

Role-based access control